

Wenyu Chiou

AI Agent Infrastructure & Evaluation Engineer

Bethlehem, PA · wec324@lehigh.edu · +1 (484) 937-2528 · github.com/WenyuChiou · linkedin.com/in/wenyu-chiou · wenyuchiou.github.io

I build AI-agent and research infrastructure — MCP pipelines, multi-agent orchestration, and agent harnesses — and I study when LLM agents can be trusted to stand in for real people, using behavioral ground-truth data I collected myself.

— SELECTED SYSTEMS

research-hub MAINTAINED OPEN-SOURCE TOOL

An AI-operable research workspace for Zotero, Obsidian, and NotebookLM — an AI assistant searches, ingests, and syncs papers through one CLI, MCP server, and REST API.

PyPI [research-hub-pipeline v1.1.1](#) · listed in [Awesome MCP Servers](#) · CI (3 OS × 4 Python)

ai-research-skills MAINTAINED OPEN-SOURCE TOOL

A 15-skill catalog that first vets whether a research idea is worth doing — open, a contribution, feasible — before any downstream work runs.

5-plugin marketplace · cross-runtime SKILL.md contract · MIT

fable-method-harness MAINTAINED OPEN-SOURCE TOOL

A portable operating harness that keeps AI-assisted work cost-aware, honest about failure, and auditable at every step.

benchmark-measured · validation CI · Claude + Codex verified

agent-collab-skills & codex-delegate MAINTAINED OPEN-SOURCE TOOLS

Multi-agent orchestration and delegation that verify their own work from git state instead of trusting the agent's report; a regression test pins a real upstream stdin hang.

CI on Ubuntu and Windows · contract + failure-mode tests

WAGF — Water Agent Governance Framework IN PREPARATION FOR RELEASE

A general framework for governed LLM agents, demonstrated in human–water simulation, that keeps coupled behavioral outcomes stable and physically sensible.

domain-rule + behavioral-theory constraints · retry feedback · audit logs

FLOODABM RESEARCH PROTOTYPE, ARCHIVED

The empirical foundation for my LLM-agent validity work: a 52,141-household simulation calibrated on a survey I designed and fielded, validated against real insurance claims — the testbed for whether LLM agents reproduce real human decisions.

937-household survey · LLM-agent substitution (WRR 2026) · NFIP mechanics · OpenFEMA-validated

— COMMUNITY & EDUCATION

awesome-agentic-ai-zh — a trilingual (zh-TW / zh-Hans / English) 7-stage roadmap for building agentic AI, with CI-enforced locale parity and external contributors. 4.5k+ GitHub stars, 588 forks (July 2026).

— OPEN-SOURCE CONTRIBUTIONS

- 10 merged pull requests in third-party open-source projects, including a root-caused engine-selection bug fix with a regression test that fails on main (BuilderIO/agent-native ×4) and a Windows file-permissions security fix (langchain-ai/openwiki).
- Taiwan-market data features (ZhuLinsen/daily_stock_analysis ×3); UTF-8 / East-Asian-width display fix (Nanako0129/coralline); documentation fix (NVIDIA/skills).

— PUBLICATION

Yang, Y. C. E., & Chiou, W. (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111. doi.org/10.1029/2025WR042111 Published · second author of two.

— EDUCATION

Ph.D., Civil & Environmental Engineering, Lehigh University	2024 – present · expected 2028
M.S., Hydrological and Oceanic Sciences, National Central University	Sep 2021 – Jun 2023
B.S., Earth Sciences, National Central University	Sep 2017 – Jun 2021

— SKILLS

AI-agent & research infrastructure: MCP servers & tooling · LLM-agent orchestration · multi-agent collaboration · agent governance & validation · research-workflow automation (Zotero / Obsidian / NotebookLM) · local model deployment (Ollama)

Programming & tooling: Python · testing & CI · Git · cross-platform (Windows / POSIX) engineering

Modeling & risk analytics: structural equation modeling (SEM) · Bayesian calibration · NFIP mechanics · exceedance-probability curves · coupled ABM × catastrophe modeling